

**Review of the SETI Post-Detection and Reply Protocols:  
Current Actions and Developments**

Leslie I. Tennen, Esq.\* and Dr. Duncan Forgan\*\*

\* *Attorney, Law Offices of Sterns and Tennen, 9524 West Camelback Road., Suite C-130, Number 347, Glendale, Arizona 85305 U.S.A., [ltennen@astrolaw.com](mailto:ltennen@astrolaw.com)*

\*\* *Centre for Exoplanet Science, SUPA, School of Physics and Astronomy, University of St Andrews, North Haugh, St Andrews, UK [dhf3@st-andrews.ac.uk](mailto:dhf3@st-andrews.ac.uk)*

***Abstract***

The International Academy of Astronautics, acting through its committee on SETI, prepared the “Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence” to guide the procedures for verification and announcement of the confirmed detection of extraterrestrial intelligence. These SETI Post-Detection Protocols were followed by the “Draft Declaration of Principles Concerning Sending Communications with Extraterrestrial Intelligence.” The Protocols have been approved and endorsed by institutions and individuals concerned with the search for extraterrestrial intelligence, including the United Nations Committee on the Peaceful Uses of Outer Space.

In the quarter century following the preparation of the Protocols, the discipline of SETI has seen significant advances in instrumentation and observational techniques, more sophisticated theoretical calculations and arguments, and a growing awareness of how to exploit astronomical data from a wide range of wavelengths for signs of intelligence. In addition, revolutionary changes have occurred in the forms of media and information dissemination on which the original Protocols were based.

The SETI Protocols were revised in 2010, however, the revisions have not been adequate to keep pace with SETI’s recent scientific and technological advances. Moreover, the 2010 revisions did not address the transformation in information dissemination, away from print and other traditional forms of mass communication to the ubiquitous social media. Recent studies have shown a clear need to re-examine the Protocols to consider both the advances in science and technology as well as the new forms of information distribution, which can place the credibility of SETI and the safety of researchers at risk if not dealt with appropriately.

This paper reports on the progress of the review of the Protocols which was initiated by the IAA SETI Permanent Committee during the 68th International Astronautical Congress in Adelaide in September 2017. The current status of this review is discussed, together with the methodology employed, including efforts to engage in a wide-ranging consultation with the SETI community, the international scientific community, and the general public. Activities relating to this review during COSPAR 2018 are discussed. Additional efforts are identified to facilitate a frank, open dialogue, that ensures as comprehensive a review as possible with the goal of achieving widely accepted, clearly articulated SETI Protocols that will be universally respected.

## I. Introduction

In 1989, The International Academy of Astronautics (IAA), acting through its Committee on the Search for Extraterrestrial Intelligence (now known as the IAA SETI Permanent Committee or SETI PC), prepared the “Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence”<sup>1</sup> to guide the procedures for verification and announcement of the confirmed detection of extraterrestrial intelligence. These SETI Post-Detection Protocols were followed by the “Draft Declaration of Principles Concerning Sending Communications with Extraterrestrial Intelligence.”<sup>2</sup> These Protocols have been approved and endorsed by institutions and individuals concerned with the search for extraterrestrial intelligence, including the United Nations Committee on the Peaceful Uses of Outer Space.<sup>3</sup>

In the quarter century following the preparation of the Protocols, revolutionary changes have occurred in the forms of media and information dissemination on which they were based. In addition, the discipline of SETI has seen significant advances in instrumentation and observational techniques, more sophisticated theoretical

calculations and arguments, and a growing awareness of how to exploit astronomical data from a wide range of wavelengths for signs of intelligence. The Protocols concentrated on SETI conducted by the classic radio method, but significant advancements have been made in optical SETI, searching, *inter alia*, for pulsed, narrowband, collimated optical light from stars;<sup>4</sup> artifact SETI looking for signs of technology, including waste heat or solar system probes;<sup>5</sup> the possibility of detecting megastructures in exoplanet transit light curves;<sup>6</sup> and data mining of public archives.<sup>7</sup>

The SETI Protocols were revised in 2010,<sup>8</sup> however, the revisions have not been adequate to keep pace with SETI’s recent scientific and technological advances. Moreover, the 2010 revisions did not address the transformation in information dissemination away from print and other traditional forms of mass communication to the ubiquitous social media. Recent studies have shown a clear need to re-examine the Protocols to consider both the advances in science and technology as well as the new forms of information distribution, which can place the credibility of SETI and the safety of researchers at risk if not dealt with appropriately.<sup>9</sup>

---

<sup>1</sup>  
<http://iaaweb.org/iaa/Scientific%20Activity/setideclaration.pdf>

<sup>2</sup>  
<http://iaaweb.org/iaa/Scientific%20Activity/setidraft.pdf>

<sup>3</sup> (Michael A. G. Michaud, John Billingham and Jill Tarter, “A Reply from Earth?”, *Acta Astronautica*, Vol. 26. No. 14, pp. 295-297 (1992).

<sup>4</sup> Tellis NK and Marcy GW (2017) *A Search for Laser Emission with Megawatt Thresholds from 5600 FGKM Stars*. ApJ, submitted, arXiv: 1704.02535.

<sup>5</sup> Isaacson H, Siemion APV, Marcy GW, Lebofsky M, Price DC, MacMahon D, Croft S, DeBoer D, Hickish J, Werthimer D, Sheikh S, Hellbourg G and Enriquez JE (2017) *The breakthrough listen search for*

*intelligent life: target selection of nearby stars and galaxies*. PASP 129, 975, pp. 054501.

<sup>6</sup> Wright JT (2017) *Exoplanets and SETI*. Invited Review, arXiv:1707.02175.

<sup>7</sup> Forgan D, Wright J, Tarter J, Korpela E, Siemion A, Almar I, Pirolat E (2018). *Rio 2.0: revising the Rio scale for SETI detections*. International Journal of Astrobiology 1–9. <https://doi.org/10.1017/S1473550418000162>

<sup>8</sup>  
<http://iaaweb.org/iaa/Scientific%20Activity/setideclaration.pdf>

<sup>9</sup> Forgan, D and Scholz, A (2016) #FoundThem 21<sup>st</sup> Century PreSearch and PostDetection *SETI Protocols for Social and Digital Media*; Sterns & Tennent, *SETI, Metalaw and Social Media*, in PRIVATE

The subject of conducting a review of the Protocols was discussed during the meeting of the SETI PC during the 68<sup>th</sup> International Astronautical Congress in September 2017, and the present authors have been tasked by the SETI PC with revising both the post-detection and reply protocols. This paper outlines the activities which have taken place in the past year, including presentations to COSPAR and others, the development of the strategy to conduct a wide-ranging review of the current protocols, and the identification of certain specific provisions which are in need of revision. The goal is to ensure as much as possible that the concerns and viewpoints of all those interested are considered in this process leading to the preparation of a revised document which will achieve broad based acceptance and adherence.

## **II. Methodology**

The authors have engaged in various activities in furtherance of the revisions, beginning with clarification and refinement of the scope of their charge and the manner in which the process will proceed. It was agreed that the consultation process should be open and include consideration of the views and concerns of professionals engaged in SETI as well as the broader international scientific community and the public. A strategy document detailing the consultation process was drafted which is discussed in detail below.

Reports on this strategy document, in the form of power point and oral presentations, were made by the authors to the 42<sup>nd</sup> COSPAR Assembly in Pasadena in July 2018. A presentation also was made earlier this year to the UK SETI Research Network.

These presentations generated substantial feedback from members of the audience. The presentation to COSPAR, in particular, was part of the session dedicated to “Searches for Advanced Extraterrestrial Life: Signatures, Strategies and Technologies,” and many attendants expressed their responses. In addition, other comments and suggestions have been received from interested persons apart from conference presentations. Many of these comments are reflected in the sampling of issues identified in Part II, below.

### *2.1 Consultation Process*

The authors were guided in the design of the consultation process by the Consultation Principles published by the UK government in 2016:

- Consultations should be clear and concise
- Consultations should be purposeful
- Consultations should be informative
- Consultations are only part of a process
- Consultations should last for a proportionate length of time
- Consultations should be targeted
- Consultations should take account of groups being consulted
- Consultations should facilitate scrutiny
- Responses to consultations should be disseminated in a timely fashion

As noted above, it was agreed that the consultations should not be limited to the professional SETI community. Thus, it was determined that a two-stream approach should be implemented, targeted to

professionals and the public, with the post-detection and reply protocols considered in separate sections of each stream.

### Stream 1: Experts in SETI

In the first stream, expert opinions will be sought from academics, legal professionals, civil servants and workers in other sectors directly related to SETI, as well as experts and professionals not directly working on SETI topics, who nevertheless have valuable contributions to make. Emphasis will be placed on the need to address modern concerns such as social media and personal protection - indeed, new classes of experts and professionals exist in these domains that did not exist in 1989, or to a degree even in 2010.

In order for the revision process to be effective, there must be broad based support from contributors to this stream to endorse the final form of revised protocols and apply them wherever possible. While the SETI community is in relative agreement about the general parameters of the Post-Detection Protocol, there is wide disagreement regarding the Reply Protocol, including whether it should even be part of these revisions. These issues are further discussed below, and it is expected that there will be heated discussions!

### Stream 2: The General Public

The SETI Protocols are written with all the people of Earth in mind. Indeed, the key intent is to ensure an honest, effective flow of information regarding SETI activity from practitioners to the general public. The Protocols can therefore be seen as a compact between those who search for extraterrestrial intelligence and the global community. Thus, it is essential that

- Individuals and governments are aware of the Protocols' existence, and
- Practitioners and funding entities agree that the Protocols represent ethical and legal conduct and best practices for SETI

A full public consultation will raise the Protocols' profile and give the general public an opportunity to participate meaningfully in the review. As a result, the public can therefore hold both governing bodies and SETI practitioners to account, as appropriate. It also allows for the expression of experiences and beliefs that are not generally reflected in the SETI community, and as such will give the revised Protocols expanded perspective and increased legitimacy.

The need for two streams in this consultation is hopefully clear. SETI experts and the general public will approach the Protocols from quite different perspectives, and apply quite heterogeneous knowledge sets to analyse, examine and comment on their content. Taking a two-stream approach allows us to give appropriately designed fora for different sets of stakeholders to give their opinions.

## 2.2 *Form of the Consultation*

Given the breadth and scope of the consultation, the process will need to be conducted using online tools which are available through customary commercial sources. The selection of specific tools to be employed is dependent on the availability of funding and other support. The possibility of coordinating the consultation with a major university is being explored.

It is anticipated that the process will entail a dedicated website, advertising the public consultations for each Protocol (Stream 2). The Protocols in their current form will be republished in PDF format on this website for reference, as well as links to selected relevant position papers and articles. Experts in the SETI community will be invited privately to contribute to Stream 1 on a closed system.

Both streams will undergo the same consultation procedure, which will begin with a brief survey noting:

- Their interest/experience in SETI, and
- Their prior awareness of the protocols

Contributors to both streams will then be invited to submit:

- Their specific comments on the current text of the Protocols, and
- Their expectations/desires for the next version of the Protocols

The website will be structured to permit comment to specific provisions, with certain issues identified as discussed below. In order to promote clarity and prevent the process from becoming unwieldy, individual comments will be limited in the number of words or characters, in an amount to be determined. Nevertheless, any limitation will be reasonable in length to permit respondents to express their concern regarding a specific issue. Moreover, respondents will be able to post comments to as many provisions of the Protocols as they desire and can post about distinguishable concerns to any specific provision in separate comments. Wherever possible, a transparent record will be kept of all contributions to the consultation.

## *2.3 Response to the Consultations*

The consultations will be open for a fixed period, such as six months, after which time a report will be prepared summarising:

- The nature of the consultation,
- Statistics on the consultation's reach and general level of engagement,
- The opinion of both streams regarding the current text of the Protocols,
- Recommendations from both streams for the next version of the Protocols

## *2.4 Activities Running Parallel to the Consultations*

The public consultation presents us with a rare opportunity to engage the wider public in SETI and in thinking about a post-detection Earth. In addition, the entire SETI community will be encouraged to reflect on its actions and principles. It is envisioned that many SETI practitioners will use the consultation as a springboard for developing public outreach projects, journal articles, software and tools, academic conferences, debates and other activities that advance a variety of aims around SETI.

## *2.5 Timeline of Activity*

The consultation can be divided into several key stages:

### *2.5.1 Develop and structure the consultation*

After determining the funding and other support available, including the above-mentioned possibility of collaboration with a major university, an appropriate two-stream consultation platform website will be developed in coordination with the IAA for attachment to or incorporation within the

IAA web domains. A press release will be prepared also in coordination with the IAA for wide distribution to announce the consultation.

#### *2.5.2 Announce the Consultation*

The press release announcing the consultation will invite the public to participate in Stream 2. At the same time, private invitations will be circulated for Stream 1 through appropriate channels, such as the membership of the IAA and IISL, representatives to COPUOS, researchers at organizations including the SETI Institute and Breakthrough Initiatives, UKSRN etc.

#### *2.5.3 Collect Consultation Responses (~ 6 months)*

During the consultation period, participation will be encouraged by advertising and promoting the existence of the website and conducting any requested media interviews. The SETI community will be encouraged to write articles and to include the subject of the revisions in conferences, debates and public outreach projects.

#### *2.6 Assimilate Consultation Responses & Publish Report*

Once the consultation is concluded, the authors will assimilate the sum total of responses and write a report for publication in, *e.g.*, *Acta Astronautica*. The report will be given to the officers of the SETI PC for comment before submission. This report will contain a condensed set of recommendations for revision from both streams. Once published, a press release will be made announcing the publication of the report, which should be made freely available on the IAA website. Further comments from both streams regarding the report will be welcomed. The process will

follow the approach for the original consultations, but with a substantially reduced period for submission of comments.

#### *2.7 Redraft the Protocols*

Based on the conclusions of the above report, and any further comments received, the authors will prepare a redraft the Protocols. This redraft will then be submitted to the IAA SETI PC for placement on the agenda at an upcoming meeting, at which time the Committee will discuss the document and either approve the revisions or articulate specific objections thereto. If necessary, additional modifications may be made to satisfy the objections, with the goal of obtaining approval of the document by consensus of the Committee if possible. If consensus is achieved, the document will then be submitted to the IAA Board of Trustees for approval, and then to the other organizations that approved the original Protocol, *e.g.*, IISL, IAF, and COPUOS etc. In the event consensus cannot be achieved, the status of the document, including whether it should be subject to approval by majority vote of the Committee, will be up to the Committee to determine.

### **III. Survey of Issues**

The foregoing discussion has been devoted to describing the process to be employed in the review of the Protocols. To be productive and effective, this process must be open, receptive and conducive to the free exchange of ideas. As such, the general and specific directions that the conversation will take are to be determined, and those determinations will be shaped by the submissions and interventions of the participants. It is anticipated that there will be topics, themes and nuances leading into issues that are as of yet unidentified, and it is

hoped that the process to be employed will be conducive to as wide-ranging and comprehensive a discussion as possible. Thus, some areas for examination and comment can be identified and suggested, while making it clear that they are offered only for the purpose of initiating the conversation, and not as a limitation thereof.

A threshold issue is to identify who are to be the parties to the Protocols. The Preamble to the Declaration of Principles identifies the parties as “individuals and institutions participating in the scientific Search for Extraterrestrial Intelligence (SETI).” This is very imprecise, for example are individuals working for or funded by an institution that is a signatory to the Protocol thereby “*ex officio*” parties to the document, especially if the employment contract or funding documentation was silent on the matter? The qualifying criterion of “participating in scientific SETI” may need refinement, to clarify the level of participation, and whether it needs to be of a professional as compared to an amateur character. Another question is whether the Protocols should be open to governments through the appropriate agency/office.

The record of who is a party to the Protocols is inadequately dealt with in the existing documents. Although there are references to the SETI Committee maintaining such a record, as a matter of practice there is not an up-to-date roster of individuals and institutions formally subscribing to the Protocols. As a technical matter, there are no signatories as such. The original Declaration of Principles was drafted by the SETI Committee, which was comprised of many of the individuals active in the field, and representatives of almost all of the institutions involved in SETI at the time. Thus, when approved by the Committee, the Declaration of Principles had the express

approval of those individuals, and the implicit approval of those institutions, and there was no formal document bearing the signatures of these persons. These circumstances were largely unchanged with the preparation and approval of the Reply Protocol. The 2010 revisions to the Declaration of Principles followed a similar path, although the meeting of the Committee which approved the revisions was attended by a relatively small number of members and thereby lacked the level of consensus achieved by the original document. It is anticipated that the discussions will include consideration of whether there should be an official list of parties to the Protocols, and if so, some details suggested as to what should be listed, when, and by whom.

The purposes of the Protocols as expressed at the time of the Declaration of Principles were: “to declare our commitment to conduct this search in a scientifically valid and transparent manner and to establish uniform procedures for the announcement of a confirmed SETI detection.” There has been a third purpose inherent in the Protocols, namely, to establish procedures to aid in the confirmation process. The SETI community has a unique responsibility to serve the public in the dissemination and understanding of information, particularly in the event of confirmed contact. Since the preparation of the original Declaration of Principles, and even since the time of the revisions in 2010, the Internet, and social media in particular, have sparked a revolution in the ways in which news and information is disseminated. This revolution has profound implications for the Protocols, both in regard to information released to the public, as well as for the safety and security of the individuals involved in the dissemination of information, and especially those involved in the event of a confirmed discovery of ETI. In this regard, the

discussion should evoke comments on how to protect individuals from hazards such as “doxxing” and account hacking.

The Protocols were drafted in a manner designed to achieve a broad consensus as a scientific endeavor across political philosophies. As a result, the terminology tended to be imprecise, leading to uncertainty in application. For example, Principle 1 provides that SETI experiments will be conducted “transparently,” and practitioners will be “responsive” to news and other media. What are the limits to transparency? Does it mean all data is open source as soon as possible? What about open source software? And what does “responsive” mean? Have relations with the media in practice, for example regarding the “interstellar visitor” Oumuamua, been conducted in conformity with these concepts?

Principle 2 concerns the handling of candidate evidence. Among its provisions are that the discoverer of potential evidence will collaborate with other investigators to confirm the detection, whether or not such other investigators are signatories to the Declaration of Principles. Principle 2 also provides that there is no obligation to disclose verification efforts while they are underway. There seems to be a tension between transparent conduct and secrecy in verification. Verification will be 99% of the effort in any *bona fide* ET detection, and it has been argued that there should be full transparency from start to finish during verification.<sup>10</sup> And what about a premature disclosure, and what if that premature disclosure turns out to be false? The disclaimer of any obligation to disclose pending verification efforts may be the most contentious provision in the whole document. A close second may be the

express inclusion of non-signatories in the confirmation process. It is not the inclusion of such persons that is the concern, rather it is the lack of any concomitant obligation to ensure that such non-signatories will respect the confidentiality of the process, at least insofar as the Declaration of Principles recognizes and preserves to the discoverer the right to make the public announcement of the confirmed detection of ET. The parameters of the interplay between confidentiality and transparency are to be determined.

The Protocols proceeded from the perspective that the discovery of an ETI most likely would be as a result of the detection by the classic radio method. Nevertheless, other methods identified above such as optical SETI may benefit from specific consideration in the Protocols. The Protocols should allow for the appropriate protection of any medium, format or other mechanism by which a detection is confirmed, just as they now mention the protection of appropriate frequencies by the World Administrative Radio Council of the International Telecommunication Union for a detection by electromagnetic signal.

Principle 3 relates to the disclosures that are to be made by the discoverer of a confirmed detection. Both the role of social media, and the interrelationship of the Rio scale to the Protocols, undoubtedly will be the subject of extensive discussions, as will be the organizations and entities which should receive a formal report, such as the United Nations, the IAU, and the other organizations which approved the original Declaration of Principles. Other questions to be considered include what restrictions can be expected to be imposed on the discoverer by an employer institution or one that

---

<sup>10</sup> Forgan & Scholz, *supra*; Forgan et al., *supra*.

provides funding? Can the information be considered proprietary? What obligation does the discoverer have to inform certain individuals, organizations, or his or her own government, prior to the public announcement, and what if the government sequesters the information? How would parties outside the jurisdiction of the government but with relevant knowledge be impacted by the sequestration?

The subjects and issues identified above are concerned with the time period of verification leading to and including the announcement of a confirmed detection. The aftermath of a confirmed detection is briefly referenced in the Declaration of Principles, while the Reply Protocol specifically addresses the sending of a message to an ETI. The Protocols refer to the former “Post-Detection Task Group under the auspices of the IAA SETI Permanent Study Group.” The PSG has been restructured since the 2010 revisions, and the status of any post-detection group is uncertain. While an examination of the structure of the SETI Permanent Committee is not part of the Protocol review, the existence and functions of any post-detection group is within the expected scope of the discussion.

The Reply Declaration broadly addressed the question of sending messages to an ETI, both pre-and post- confirmation of detection. The former, METI, has been the subject of heated debate and division within the SETI community,<sup>11</sup> and the latter, a response to a confirmed detection, also is fraught with controversy. Although the authors do not seek to re-ignite smoldering and contentious arguments, it is anticipated that these matters may provoke considerable commentary. It is proposed that the survey

include consideration of both METI and a response to a confirmed detection, and that the decision whether or not to include either or both in the revisions to the protocols deferred pending the content of the consultations, including an assessment of whether there is reason to believe that a consensus can be achieved on applicable provisions.

It is expected that the specific issues to be discussed will include: should there be a global response to a confirmed detection, how will that be developed, and which states, entities, organizations, and individuals will be permitted to participate in the process. How can it be established that a global response is a *bona fide* official response distinguishable from any other messages that may originate from Earth; what prior consultations and/or consent, if any should be obtained to the sending of a non-global official message; and what are the ramifications of failure to obtain such consultations/consent, especially for non-state actors? Should there be any formal catalogue and monitoring of METI transmissions?

It is not intended that the consultation will include a discussion of what should be the content of a global response or METI transmission. Any response will, of course, largely depend on the context of the confirmed detection, whether such detection includes substantive content, and the then extant political climate. It is intended that the consultations will include consideration of the use of social media both for public input and for message composition. It also is expected that the discussion will include consideration of a long-term institutional framework for an exchange of communications in the event of a confirmed

---

<sup>11</sup> See generally The METI Debate, JBIS Vol 67 No 01 – January 2014.

detection and a response thereto. In this regard, there may be some discussion of the metalegal principles that will influence the relationship between the two intelligences.

Finally, it is anticipated that the discussion will consider what are to be the consequences of an individual or institution's failure to comply with their obligations as set forth in the Protocols. Are the Protocols to be mere guidelines or are they to be more in the nature of standards or a code of conduct that are voluntarily binding. The current documents are silent as to consequences or remedies for non-compliance. The issue of enforcement is perhaps one of the most difficult areas in which to seek consensus, and the expected conversation on this topic will be both interesting and illuminating.

#### ***IV. Concluding Remarks***

The Search for Extraterrestrial Intelligence has undergone a technological and computational renaissance, which has outstripped our thinking on the legal and policy issues surrounding the endeavor.

The detection and confirmation of extraterrestrial intelligence (if it occurs) is likely to send significant shock waves through many aspects of human society - scientific, cultural and political. The risks to society (and the individuals participating in the discovery) are potentially great. It is therefore important that we achieve consensus on a robust, respectful and dignified set of Protocols for discoverers to follow, and that these Protocols are well understood and agreed upon by humanity at large.

This paper outlines our efforts to bring post-detection policy up to date. With the approval of the IAA Permanent Committee

on SETI, we have begun preparations for a consultation of both the SETI community and humanity as a whole on the expected actions and behavior of SETI practitioners.

The foregoing discussion has defined the scope and parameters of the consultation, as well as identified some of the issues that the authors have considered, as well as some that have arisen from colleagues during our initial preparations. Even at this early stage, it is clear that there will be considerable debate on the finer points of the revised Protocols – this consultation will prove to be a valuable tool in mediating these disagreements.